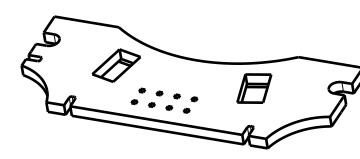
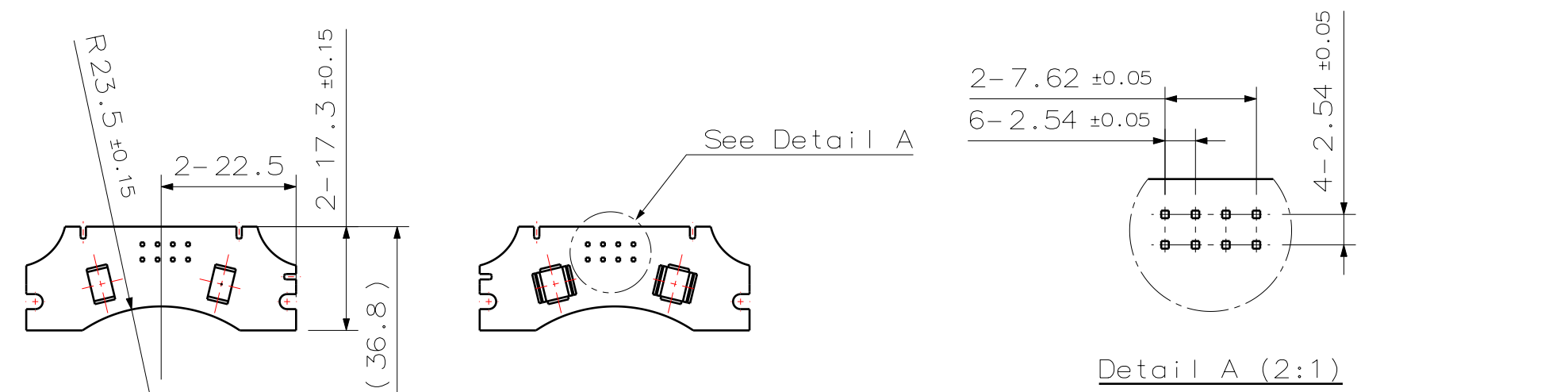


All parts which supply to LG Innotek must not contain prohibited substances including RoHS Hazardous substances and for more details refer to LG Innotek's "Manual for management of hazardous substances in Product"

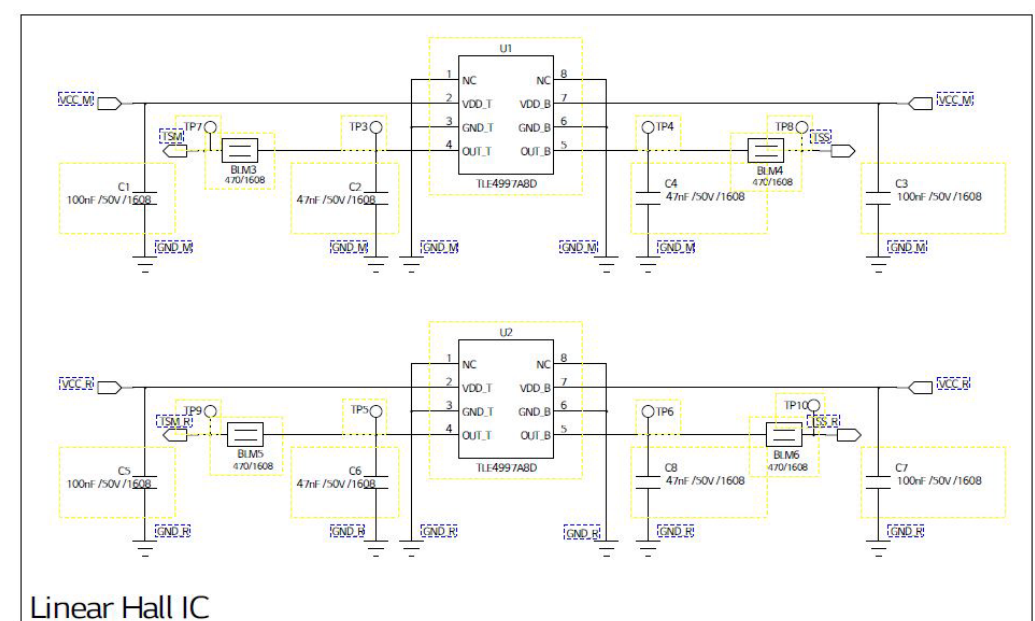
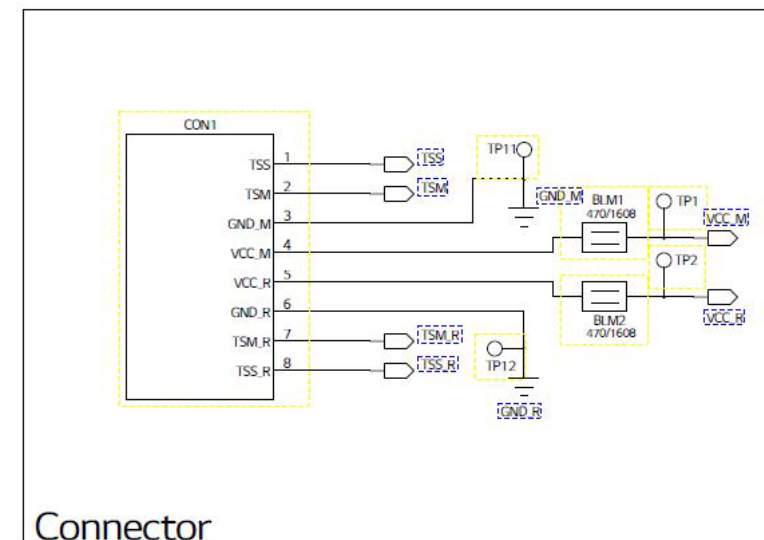
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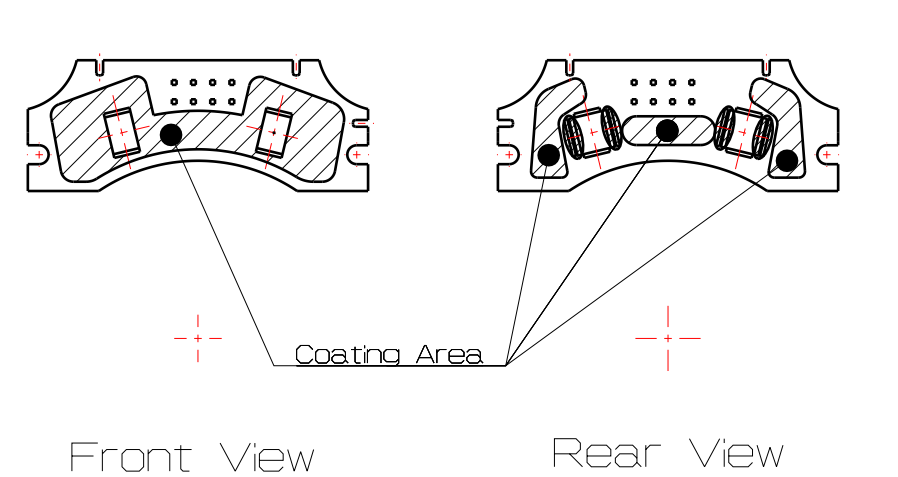
Isometric view (1:1)

- NOTE
1. This is a PCB Ass'y drawing for TOS.
 2. Regarding the detail for circuit pattern, follow the artwork.
 3. No swap error, omission and direction of the chip component.
 4. No PCB degradation, burn out, flux and alien substance after SMT.
 5. No deformation and damage at PCB.
 6. All components certificated for automotive field must be used.
 7. PCB conformal coating spec
 - 1) Conformal coating spec
 - DOW CORNING 3-1953
 - 2) Conformal coating area
 - Follow the left conformal coating area.
 - The surface mount of the circuit components and via holes
 - 3) No conformal coating area
 - Follow the left conformal coating area
 - Exception area of coating : Connector holes, Screw holes, Hall IC hole area and assembly area with case
 - 4) Conformal coating thickness : 60 ~ 150um
 8. Appearance criteria after coating follows the PQR Rev. 3
 9. Coating liquid must not spill out on the side of the PCB.
 10. All operators must wear a ESD protection bracelet, and must be equipped with ionizer in the main process(material warehouse, IC).
 11. This part must be satisfied with the following regulation.
 - 1) Meet ES-90000-01 Rev.9
 - 2) Meet ES-90000-02 Rev.6
 - 3) Meet ES-90000-03 Rev.5
 - 4) Meet ES-90000-04 Rev.7
 - 5) Meet ES-90000-06 Rev.3
 - 6) Meet MS-201-02 Rev.39
 - 7) Meet MS-915-01 <TYPE A> Rev.3
 - 8) Meet MS-941-01 <TYPE B> Rev.8
 - 9) Meet MS-941-04 <TYPE A-1> Rev.9
 - 10) Meet MS-184-01 <TYPE B> Rev.6
 - 11) Meet PQR Rev. 3
 12. PCB Ass'y bending Quantity : Max 500um/m(No crack all part)
 13. PCB Revision No. : A
 14. Solder Paste : TAMURA TLF-204-HSP
 15. All PCB Ass'y must be inspected as the following 'PCB Ass'y Test List'
 16. Torque output inspection : 50us(Frequency), 1,000times @PCB Ass'y

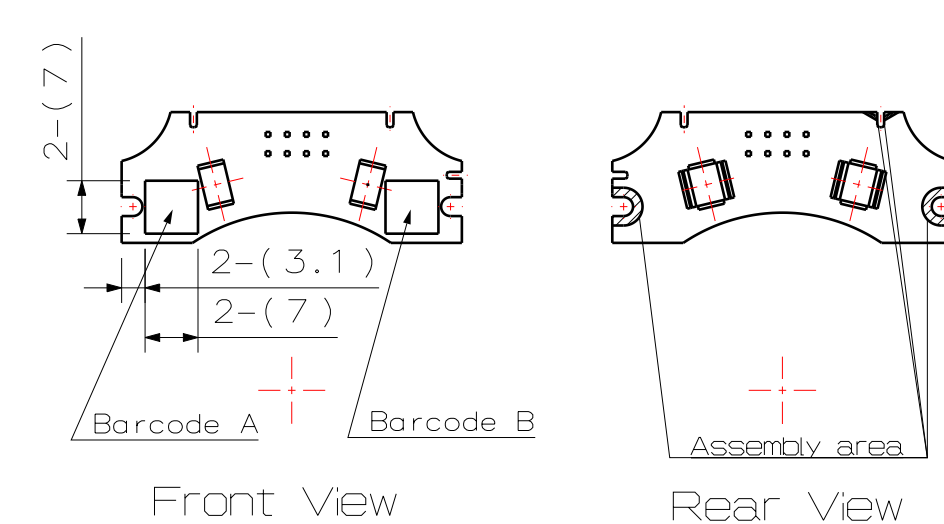
Circuit Drawing



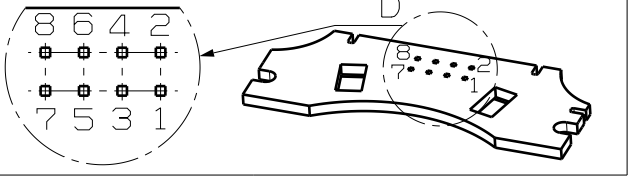
Coating Area



Barcode and Assembly area with case

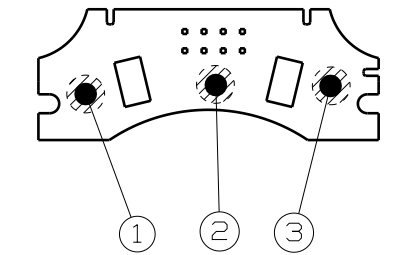


Pin Number



Pin Map

Pin Number	Name	Name
1	TSS	
2	TSM	
3	GND_M	
4	VCC_M	
5	VCC_R	
6	GND_R	
7	TSM_R	
8	TSS_R	



Deformation measurement 3points of PCB Ass'y (See Notes '12')

PCB Assy Test List

Test Item	Test Method	Acceptance Criteria
1 TOS output	Measure voltage of TSM, TSS, TSM_R, TSS_R after applying 5V_at Power A, Power B (Refer to NOTE 16)	2.5V ± 0.1V
2 TOS Output Variation	Measure voltage of TSM, TSS, TSM_R, TSS_R after applying 5V at Power A, Power B (Applying ECU Interface Circuit)	1) TSM < 7mV, TSM_R < 7mV 2) TSS < 7mV, TSS_R < 7mV 3) TSM-TSS < 7mV, TSM_R-TSS_R < 7mV (Applying ECU Interface Circuit)
3 TOS consumption current	Measure consumption after applying 5V at Power A, Power B	VCC_M : Max. 20mA, VCC_R : Max. 20mA

- Power A : TSM(T1), TSS(T2) Power Line Pin No.4(VCC_M), No.3(GND_M)
- Power B : TSM_R(T3), TSS_R(T4) Power Line Pin No.5(VCC_R), No.6(GND_R)
- TOS Hall IC output are Voltage Output before program.

DIMENSIONAL TOLERANCE	
LENGTH	±0.1
ANGULAR	±0.5°
UNLESS OTHERWISE SPECIFIED	

CHANGES	REV NO.	DATE (YY MM DD)	SIGNATURE	CHANGE CONTENTS
A	220110	J.Y.YANG		Change the coating area (EOKLM240)
B	210702	J.Y.YANG		Change the Coating Area and Note
C	210414	J.Y.YANG		Released

Q'TY	NO	PART NO	DESCRIPTION	MATERIAL	COLOR FINISH	NOTE
4	5	2CAC01466A	Samsung Electro Mechanics, Open Mode MLC 100nF/50V/1608, CL10B104K8WPN			C1,C3,C5,C7
4	4	2CAC01803A	Samsung Electro Mechanics, Open Mode MLC 47nF/50V/1608, CL10B473K8WPN			C2,C4,C6,C8
6	3	2FTC00096A	Murata, Ferrite Bead 470/1608, BLM18AG471SH1			BLM1, BLM2, BLM3, BLM4, BLM5, BLM6
2	2	2ICZ00367A	Infineon, Linear Hall IC TLE4997ABD			U1, U2
1	1	3PCR03017A	PCB, FR-4, 2Layer, Middle-Tg, Gold Plated			-
RELATED P/N		SCALE	UNIT	DESIGN	TITLE	
		THIRD ANGLE PROJECT	1:1 mm	2021.04.14	PCB Assy	
				2021.04.14	PART NO	
					MODEL	
				2021.04.14	DWG NO	
					3MAP00490A	

